

Right-Angled Triangles: Pythagoras and Trigonometry

Two sides → Pythagoras. A side and an angle → SOH CAH TOA. Label the sides from the angle first.



DR ESLAM AHMED
eliteigcse.com

THE TOPIC ON ONE MAP

Right-Angled Triangles: Pythagoras and Trigonometry

01 Pythagoras

hypotenuse shorter side is it right-angled?

02 SOH CAH TOA

find a side find an angle

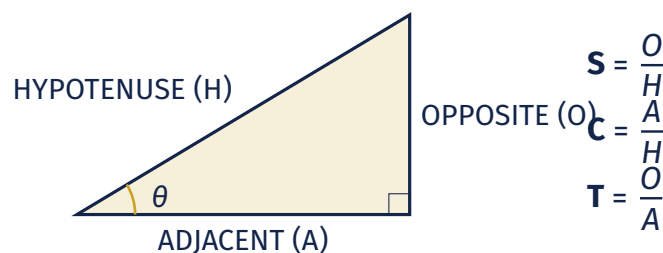
03 Elevation, depression, real life

cliff and buoy zip wire

04 Two triangles together

shared side side first, then angle

LABEL, THEN CHOOSE Name the sides from the angle you use:



01 Pythagoras

Only for right-angled triangles. Longest side: add the squares. Shorter side: subtract. Always check the hypotenuse is the longest answer.

WHEN YOU SEE → YOU DO

two sides, want the third, no angles

→ Pythagoras

want the hypotenuse

→ $c = \sqrt{a^2 + b^2}$

want a shorter side

→ $a = \sqrt{c^2 - b^2}$

“is the triangle right-angled?”

→ check $a^2 + b^2 = c^2$

a rectangle's diagonal, an isosceles height

→ split into right-angled triangles

WORKED MODEL • NOVEMBER 2025 P2H

Right angle at A, $AB = 30$ cm, area 240 cm². Perimeter?

1 $\frac{1}{2}(30)(AC) = 240 \Rightarrow AC = 16$

area

2 $BC = \sqrt{30^2 + 16^2} = 34$

Pythagoras

ANSWER $30 + 16 + 34 = 80$ cm

IDEA 01

Practise: Pythagoras

My questions first, then selected practice, then real past papers.

Q1 DR ESLAM Hypotenuse

CH 4 • TOPIC 10 2 marks

A right-angled triangle has shorter sides 7 cm and 24 cm. Find the hypotenuse.

Q2 DR ESLAM Shorter side

CH 4 • TOPIC 10 2 marks

The hypotenuse is 13 cm and one side is 5 cm. Find the other side.

SELECTED PRACTICE

Q3 **PRACTICE** Practice • Easy

CH 4 • TOPIC 10 3 marks

- 1 XYZ is a right-angled triangle.

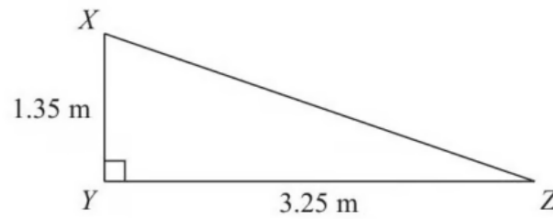


Diagram **NOT**
accurately drawn

Calculate the length of XZ .

Give your answer correct to 3 significant figures.

(3 marks)

Q4**PRACTICE**

Practice · Medium

CH 4 · TOPIC 10

5 marks

3 Here is part of a field.

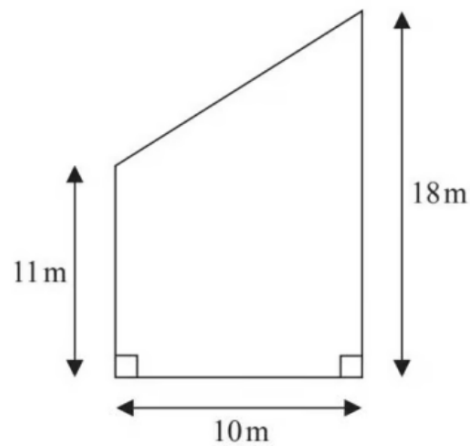


Diagram **NOT**
accurately drawn

This part of the field is in the shape of a trapezium.

A farmer wants to put a fence all the way around the edge of this part of the field.

The farmer has 50m of fence.

Does he have enough fence?

You must show all your working.

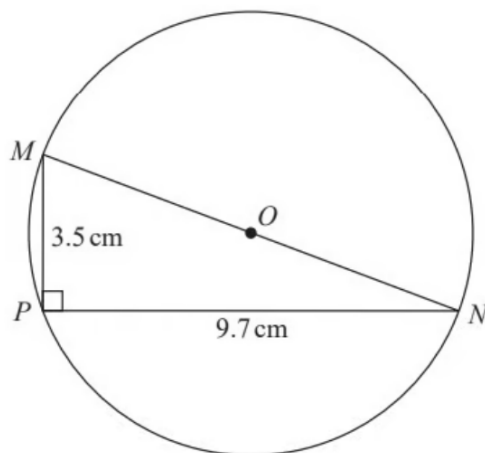
(5 marks)

Q5**PRACTICE**

Practice · Medium

CH 4 · TOPIC 10 4 marks

14

Diagram **NOT**
accurately drawn

M , N and P are points on a circle, centre O .

MON is a diameter of the circle.

$$MP = 3.5 \text{ cm}$$

$$PN = 9.7 \text{ cm}$$

$$\text{Angle } MPN = 90^\circ$$

Work out the circumference of the circle.

Give your answer correct to 3 significant figures.

..... cm

(4 marks)

Q6**PRACTICE**

Practice · Very Hard

CH 4 · TOPIC 10

4 marks

- 4** A square, with sides of length x cm, is inside a circle.
Each vertex of the square is on the circumference of the circle.

The area of the circle is 49 cm^2 .

Work out the value of x .

Give your answer correct to 3 significant figures.

(4 marks)

PAST PAPERS

Q7**PAST PAPER**

Edexcel 4MA1 • January 2020 • Paper 2HR • Q4

CH 4 • TOPIC 10 3 marks

- 4 The diagram shows a rectangle and a diagonal of the rectangle.

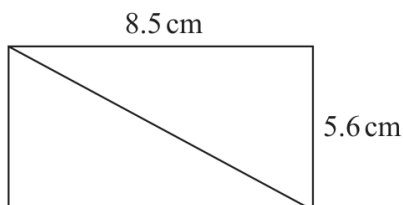


Diagram **NOT**
accurately drawn

Work out the length of the diagonal of the rectangle.
Give your answer correct to 1 decimal place.

..... cm

(Total for Question 4 is 3 marks)

Q8**PAST PAPER**

Edexcel 4MA1 · November 2021 · Paper 1H · Q2

CH 4 · TOPIC 10**3 marks**

- 2 Here is a right-angled triangle.

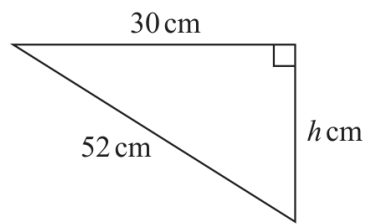


Diagram **NOT**
accurately drawn

Calculate the value of h .

Give your answer correct to 3 significant figures.

$h =$

(Total for Question 2 is 3 marks)

Q9

PAST PAPER

Edexcel 4MA1 · January 2022 · Paper 1HR · Q3

CH 4 · TOPIC 10 4 marks

- 3 The shaded shape is made using three identical right-angled triangles and a square.

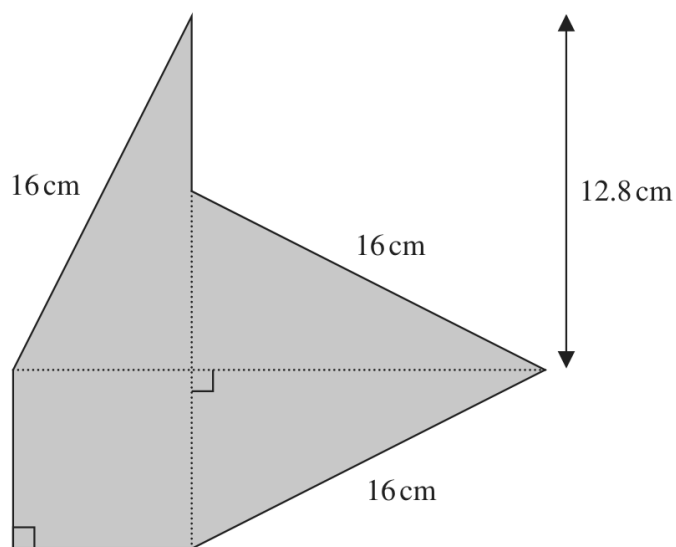


Diagram **NOT**
accurately drawn

Work out the perimeter of the shaded shape.

..... cm

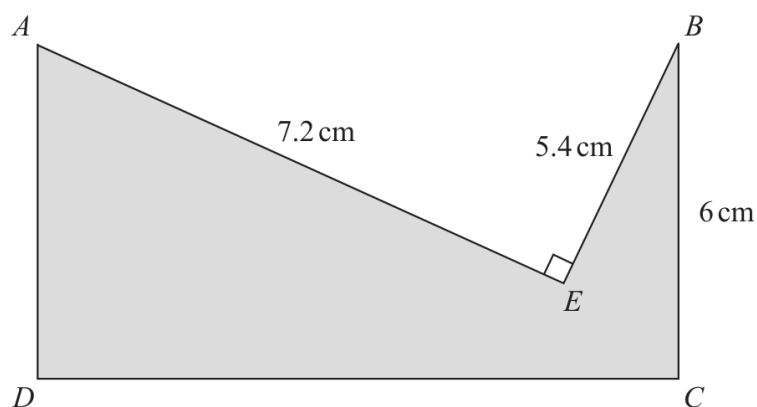
(Total for Question 3 is 4 marks)

Q10**PAST PAPER**

Edexcel 4MA1 • May 2023 • Paper 1H • Q5

CH 4 • TOPIC 10 4 marks

- 5 The diagram shows a shaded shape $AEBCD$ made by removing triangle AEB from rectangle $ABCD$



$$AE = 7.2 \text{ cm} \quad BE = 5.4 \text{ cm} \quad BC = 6 \text{ cm} \quad \text{angle } AEB = 90^\circ$$

Work out the perimeter of the shaded shape.

..... cm

(Total for Question 5 is 4 marks)

Q11 PAST PAPER Edexcel 4MA1 · May 2024 · Paper 1H · Q11

CH 4 · TOPIC 10 4 marks

11 The diagram shows a roof support.

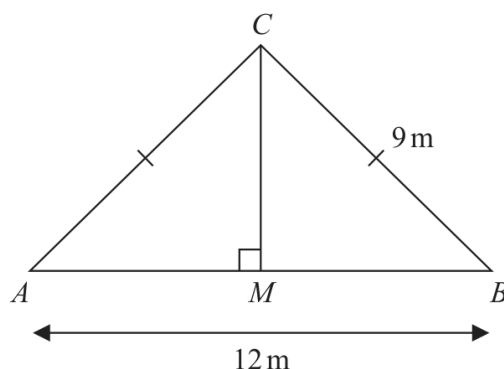


Diagram **NOT**
accurately drawn

The roof support is made from four lengths of wood, AB , AC , BC and MC

$$AC = BC = 9 \text{ m} \quad AB = 12 \text{ m}$$

$$\text{angle } AMC = 90^\circ$$

Lewis is going to buy lengths of wood to make the roof support.

The wood costs 21.50 euros per metre.

Each length of wood he buys has to be a whole number of metres.

Work out the total cost of the wood Lewis needs to buy.

Show your working clearly.

..... euros

(Total for Question 11 is 4 marks)

Q12**PAST PAPER**

Edexcel 4MA1 · November 2025 · Paper 2H · Q11

CH 4 · TOPIC 10 4 marks

11 The diagram shows a right-angled triangle ABC

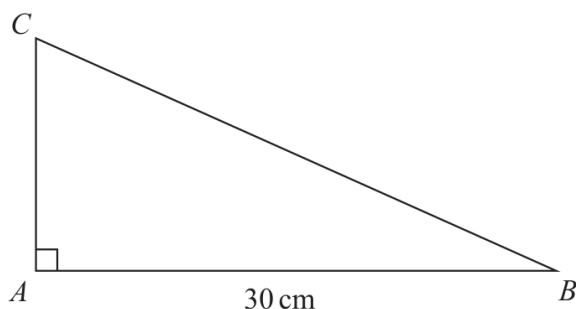


Diagram **NOT**
accurately drawn

Area of triangle $ABC = 240 \text{ cm}^2$

Work out the perimeter of triangle ABC

..... cm

(Total for Question 11 is 4 marks)

02 SOH CAH TOA

Label O, A, H from the angle. Pick the ratio that uses the two sides you have or want. To find an angle, use the inverse ($\sin^{-1}$, $\cos^{-1}$, $\tan^{-1}$).

WHEN YOU SEE → YOU DO

angle and one side, want a side

→ choose S, C or T; unknown on top: multiply; on the bottom: divide

two sides, want an angle

→ inverse function

“correct to 1 d.p.”

→ calculator in degrees

an isosceles triangle

→ split down the middle into two right-angled triangles

Label O A H

Choose SOH CAH TOA

Rearrange

Calculate

WORKED MODEL • JUNE 2025 P1HR

$PQ = 6.5$, angle $R = 24^\circ$, right angle at Q . Find QR .

1 $PQ = \text{opposite}$, $QR = \text{adjacent} \rightarrow \text{TOA}$

label

$$2 \quad \tan 24^\circ = \frac{6.5}{QR} \Rightarrow QR = \frac{6.5}{\tan 24^\circ}$$

unknown on the bottom: divide

ANSWER $QR = 14.6 \text{ cm}$

× **TRAP** Calculator in radians: $\tan 24 = -2.13$. Check the D on the screen.

IDEA 02

Practise: SOH CAH TOA

My questions first, then selected practice, then real past papers.

Q13 DR ESLAM Find a side

CH 4 • TOPIC 10 2 marks

In a right-angled triangle the hypotenuse is 10 cm and one angle is 35° . Find the side opposite 35° .

Q14 DR ESLAM Find an angle

CH 4 • TOPIC 10 2 marks

A right-angled triangle has opposite 5 cm and adjacent 12 cm. Find the angle.

SELECTED PRACTICE

Q15

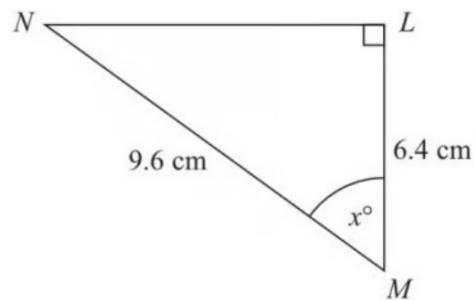
PRACTICE

Practice · Easy

CH 4 · TOPIC 10

3 marks

3

Diagram **NOT**
accurately drawn LMN is a right-angled triangle. $MN = 9.6$ cm. $LM = 6.4$ cm.Calculate the size of the angle marked x° .

Give your answer correct to 1 decimal place.

(3 marks)

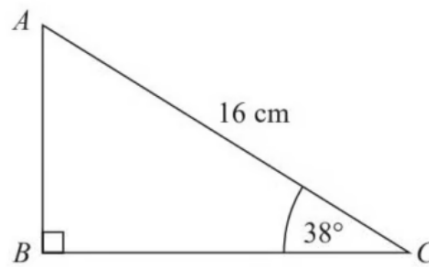
Q16**PRACTICE**

Practice · Easy

CH 4 · TOPIC 10

2 marks

8 ABC is a right-angled triangle.



Calculate the length of AB .

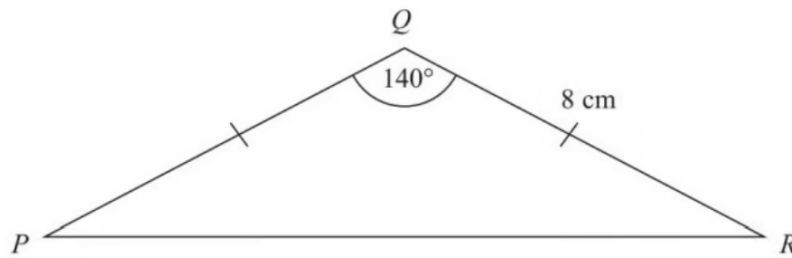
Give your answer correct to 2 decimal places.

(2 marks)

Q17**PRACTICE**

Practice · Medium

CH 4 · TOPIC 10

3 marks**10**Diagram **NOT**
accurately drawnCalculate the length of PR .

Give your answer correct to 3 significant figures.

(3 marks)

Q18**PRACTICE**

Practice · Hard

CH 4 · TOPIC 10

3 marks

10 Here is a parallelogram.

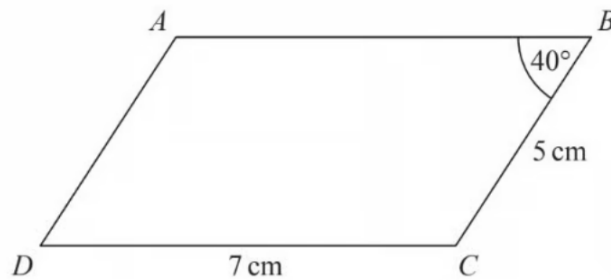


Diagram **NOT**
accurately drawn

$$DC = 7\text{ cm}$$

$$CB = 5\text{ cm}$$

Angle ABC is 40°

Work out the area of the parallelogram.

Give your answer correct to 1 decimal place.

(3 marks)

PAST PAPERS

Q19**PAST PAPER**

Edexcel 4MA1 · January 2020 · Paper 1H · Q9

CH 4 · TOPIC 10 **3 marks**

- 9 Here is a right-angled triangle.

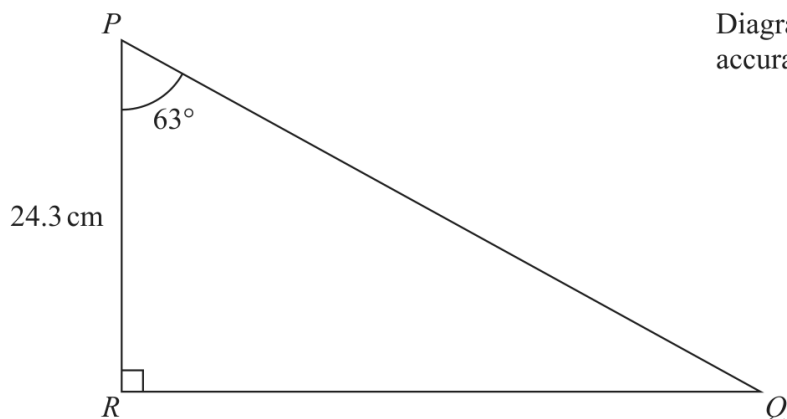


Diagram **NOT**
accurately drawn

Calculate the length of PQ .

Give your answer correct to 3 significant figures.

..... cm

(Total for Question 9 is 3 marks)

Q20**PAST PAPER**

Edexcel 4MA1 · January 2021 · Paper 1H · Q10

CH 4 · TOPIC 10 4 marks

10 Here is isosceles triangle ABC .

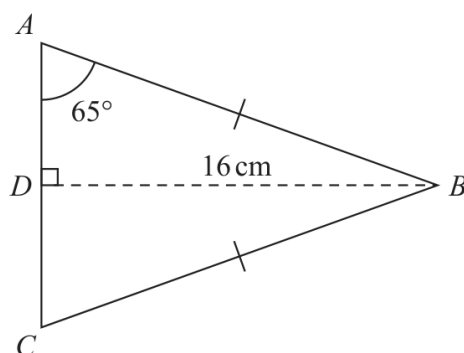


Diagram **NOT**
accurately drawn

D is the midpoint of AC and $DB = 16$ cm.

Angle $DAB = 65^\circ$

Work out the perimeter of triangle ABC .

Give your answer correct to one decimal place.

..... cm

(Total for Question 10 is 4 marks)

Q21**PAST PAPER**

Edexcel 4MA1 • May 2021 • Paper 2H • Q7

CH 4 • TOPIC 10 3 marks

7 Here is a right-angled triangle.

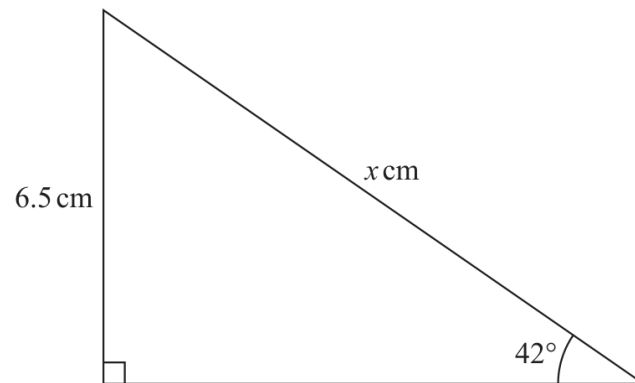


Diagram **NOT**
accurately drawn

Work out the value of x .

Give your answer correct to one decimal place.

$x =$

(Total for Question 7 is 3 marks)

Q22**PAST PAPER**

Edexcel 4MA1 · January 2022 · Paper 1H · Q3

CH 4 · TOPIC 10 3 marks

- 3 The diagram shows triangle ABC

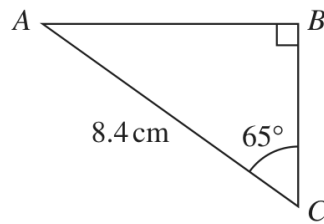


Diagram **NOT**
accurately drawn

Work out the length of the side AB

Give your answer correct to 3 significant figures.

..... cm

(Total for Question 3 is 3 marks)

Q23**PAST PAPER**

Edexcel 4MA1 · June 2022 · Paper 2H · Q6

CH 4 · TOPIC 10 **3 marks**

- 6 The diagram shows triangle PQR .

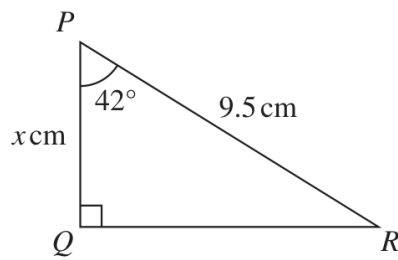


Diagram **NOT**
accurately drawn

Work out the value of x
Give your answer correct to one decimal place.

$x =$

(Total for Question 6 is 3 marks)

Q24**PAST PAPER**

Edexcel 4MA1 · June 2024 · Paper 2H · Q5

CH 4 · TOPIC 10 3 marks

- 5 The diagram shows triangle ABC

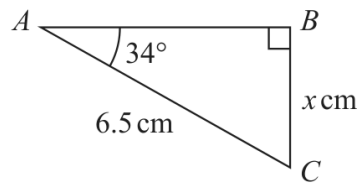


Diagram **NOT**
accurately drawn

Work out the value of x

Give your answer correct to one decimal place.

$x =$

(Total for Question 5 is 3 marks)

Q25**PAST PAPER**

Edexcel 4MA1 • November 2024 • Paper 2H • Q3

CH 4 • TOPIC 10 3 marks

- 3 The diagram shows triangle PQR

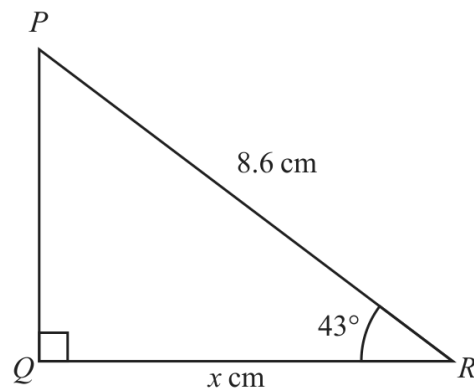


Diagram **NOT**
accurately drawn

Work out the value of x
Give your answer correct to one decimal place.

$x =$

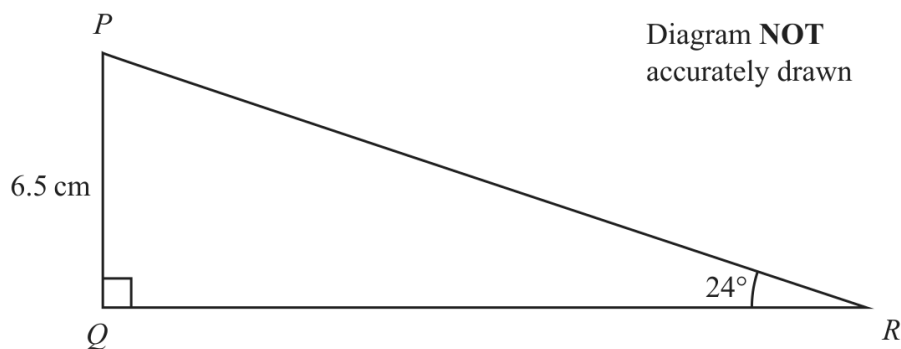
(Total for Question 3 is 3 marks)

Q26**PAST PAPER**

Edexcel 4MA1 • June 2025 • Paper 1HR • Q9

CH 4 • TOPIC 10 3 marks

- 9 The diagram shows triangle PQR



Work out the length of QR

Give your answer correct to 3 significant figures.

..... cm

(Total for Question 9 is 3 marks)

03 Elevation, Depression and Real Life

Draw the right-angled triangle with the horizontal. Elevation is measured up from the horizontal; depression down from it. They are equal (alternate angles).

WHEN YOU SEE → YOU DO

“angle of elevation from B”

→ at B, up from the ground

“angle of depression of B from A”

→ at A, down from the horizontal

two masts / posts

→ the difference in heights is the opposite side

“height of the mast on top of the cliff”

→ two triangles; subtract

WORKED MODEL • JANUARY 2022 P2HR

Masts $AB = 6.2$ m and $CD = 30.7$ m are 244 m apart. Angle of depression of B from D?

1 difference in height = $30.7 - 6.2 = 24.5$

opposite

2 $\tan \theta = \frac{24.5}{244}$

TOA

ANSWER $\theta = 5.7^\circ$

IDEA 03

Practise: Elevation, Depression and Real Life

My questions first, then selected practice, then real past papers.

Q27 DR ESLAM Elevation

CH 4 • TOPIC 10 3 marks

From 40 m away, the angle of elevation of the top of a tower is 32° . Find the height of the tower.

SELECTED PRACTICE

Q28**PRACTICE**

Practice • Medium

CH 4 • TOPIC 10 3 marks

- 6 The diagram shows a ladder leaning against a vertical wall.

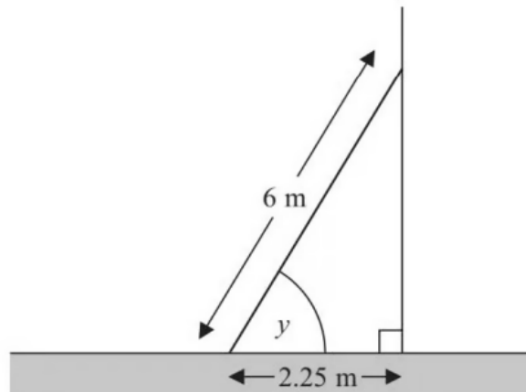


Diagram **NOT**
accurately drawn

The ladder stands on horizontal ground.

The length of the ladder is 6 m.

The bottom of the ladder is 2.25 m from the bottom of the wall.

A ladder is safe to use when the angle marked y is about 75° .

Is the ladder safe to use?

You must show all your working.

(3 marks)

Q29**PRACTICE**

Practice • Medium

CH 4 • TOPIC 10 7 marks

- 8 (a) The diagram shows the positions of three turbines A , B and C .

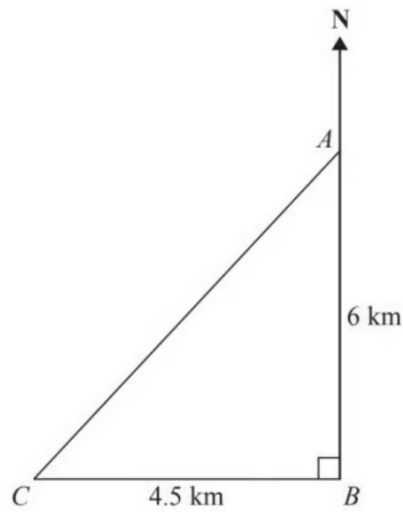


Diagram NOT
accurately drawn

A is 6 km due north of turbine B .
 C is 4.5 km due west of turbine B .

Calculate the distance AC .

(3 marks)

- (b) Calculate the bearing of C from A .
Give your answer correct to the nearest degree.

(4 marks)

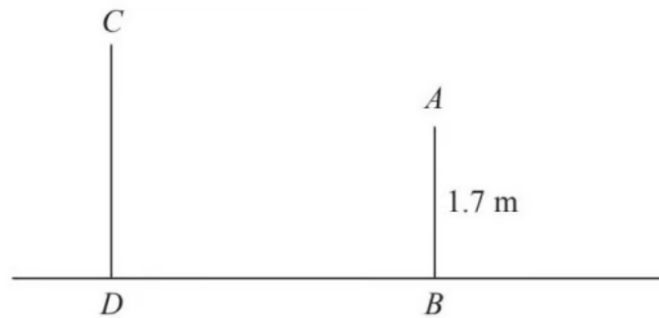
Q30**PRACTICE**

Practice · Hard

CH 4 · TOPIC 10

4 marks

- 3 The diagram shows two vertical posts, AB and CD , on horizontal ground.



$$AB = 1.7\text{ m}$$

$$CD : AB = 1.5 : 1$$

The angle of elevation of C from A is 52° .

Calculate the length of BD .

Give your answer correct to 3 significant figures.

(4 marks)

Q31

PRACTICE

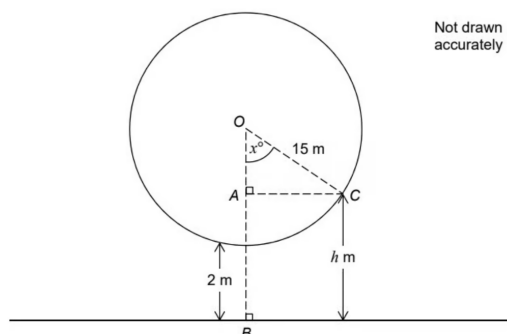
Practice · Very Hard

CH 4 · TOPIC 10 6 marks

8 (a) A Big Wheel is modelled as a circle with centre O and radius 15 metres.

The wheel turns in an anticlockwise direction.

The lowest point on the wheel is always 2 metres above horizontal ground.



C is a point on the wheel, h metres above horizontal ground.

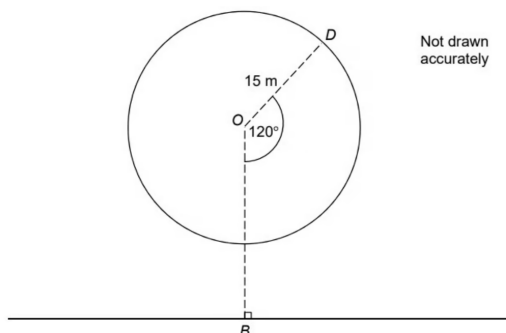
Angle $COB = x^\circ$

Show that $h = 17 - 15 \cos x^\circ$

(2 marks)

(b) D is a point on the wheel.

Angle $DOB = 120^\circ$



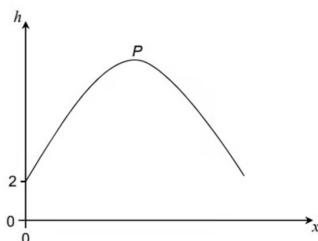
Work out the height of D above horizontal ground.

..... metres

(2 marks)

(c) Here is a sketch of the graph $h = 17 - 15 \cos x^\circ$ for one **complete** turn of the wheel.

P is the highest point on the graph.



Work out the coordinates of P .

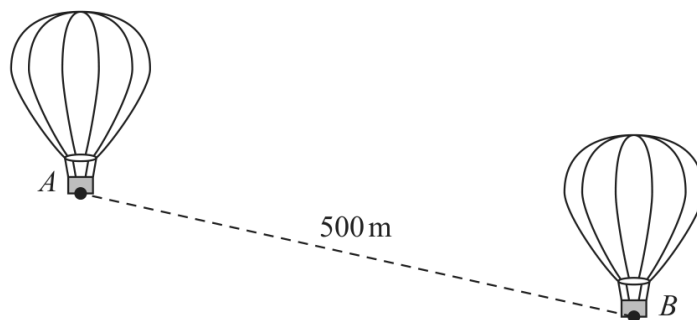
(2 marks)

PAST PAPERS

Q32 PAST PAPER Edexcel 4MA1 • January 2021 • Paper 2HR • Q13

CH 4 • TOPIC 10 3 marks

- 13 The diagram shows two hot air balloons.
 A is a point on the base of one of the balloons and B is a point on the base of the other balloon.

Diagram **NOT**
accurately drawn

The distance between A and B is 500 metres.
The angle of depression of B from A is 23°

Calculate the vertical height of A above B .
Give your answer correct to one decimal place.

..... metres

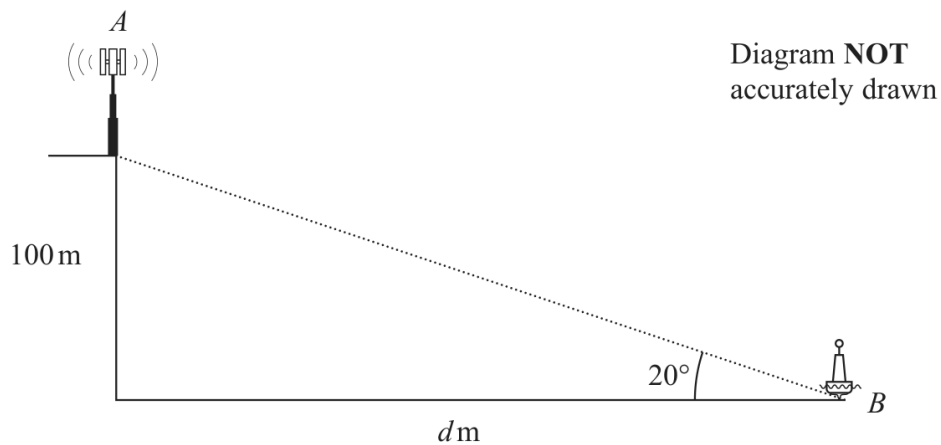
(Total for Question 13 is 3 marks)

Q33**PAST PAPER**

Edexcel 4MA1 • November 2021 • Paper 1H • Q12

CH 4 • TOPIC 10 6 marks

- 12 The diagram shows a vertical cliff with a vertical radio mast on top of the cliff and a buoy in the sea.



The height of the cliff is 100 metres.

The buoy is at the point B that is d metres from the base of the cliff.

The angle of elevation from B to the top of the cliff is 20°

- (a) Calculate the value of d .

Give your answer correct to 3 significant figures.

$d = \dots\dots\dots$

The point A at the top of the radio mast is vertically above the top of the cliff.

The angle of elevation from B to A is 25°

- (b) Calculate the height of the radio mast.

Give your answer correct to 3 significant figures.

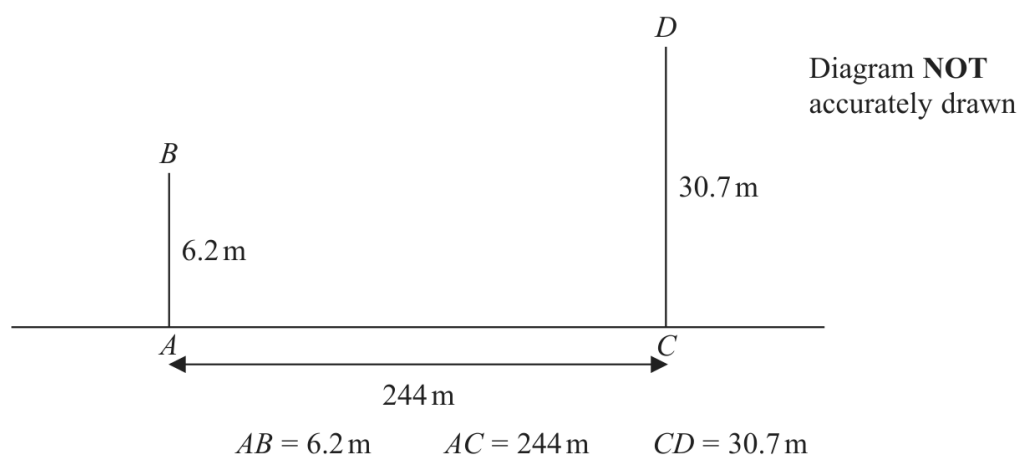
$\dots\dots\dots$ m

(Total for Question 12 is 6 marks)

Q34 PAST PAPER Edexcel 4MA1 · January 2022 · Paper 2HR · Q12

CH 4 · TOPIC 10 3 marks

12 The diagram shows two vertical phone masts, AB and CD , on horizontal ground.



Work out the size of the angle of depression of B from D
Give your answer correct to one decimal place.

○

(Total for Question 12 is 3 marks)

Q35**PAST PAPER**

Edexcel 4MA1 • January 2022 • Paper 2H • Q14

CH 4 • TOPIC 10 3 marks

- 14 A zip wire is shown as the dashed line AC in the diagram.

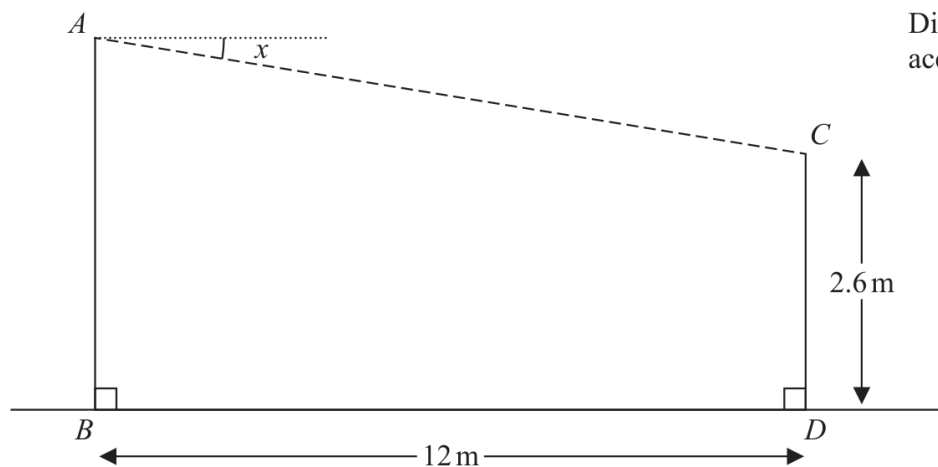


Diagram **NOT**
accurately drawn

The zip wire is supported by two vertical posts AB and CD standing on horizontal ground.

$$CD = 2.6 \text{ m} \quad BD = 12 \text{ m}$$

The zip wire makes an angle x with the horizontal, as shown in the diagram.
The design of the zip wire requires the angle x to be at least 5°

Work out the least possible height of the post AB
Give your answer correct to 3 significant figures.

..... m

(Total for Question 14 is 3 marks)

04 Two Triangles Together

Find the side the two triangles share first, then use it in the second triangle. Keep the full calculator value in between.

WHEN YOU SEE → YOU DO

two right-angled triangles joined

→ shared side first

a point on a side, “work out DC ”

→ whole – part

a trapezium or a regular polygon

→ drop a perpendicular to make right angles

“give your answer to 3 s.f.”

→ no rounding in the middle

WORKED MODEL • NOVEMBER 2024 P1H

$AB = 4250$ m, right angle at B ; angle $BAD = 47^\circ$; angle $BCA = 24^\circ$. Find DC .

$$1 \quad BC = \frac{4250}{\tan 24^\circ} = 9545.7$$

triangle ABC

$$2 \quad BD = 4250 \tan 47^\circ = 4557.6$$

triangle ABD

ANSWER $DC = 4988$ m

IDEA 04

Practise: Two Triangles Together

My questions first, then selected practice, then real past papers.

SELECTED PRACTICE**Q36**

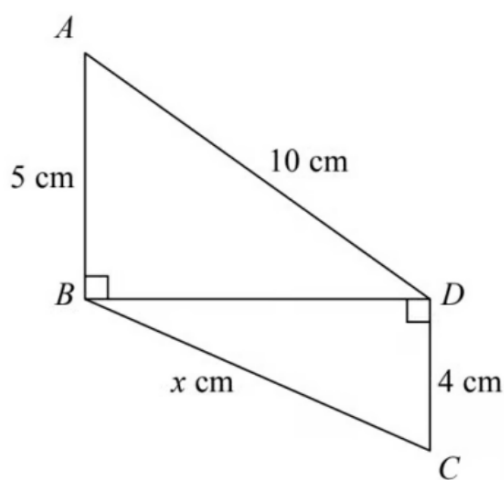
PRACTICE

Practice · Hard

CH 4 · TOPIC 10

4 marks

- 1 Triangles ABD and BCD are right-angled triangles.



Work out the value of x .

Give your answer correct to 2 decimal places.

(4 marks)

Q37**PRACTICE**

Practice • Hard

CH 4 • TOPIC 10 4 marks

- 9 The diagram shows a triangle DEF inside a rectangle $ABCD$.

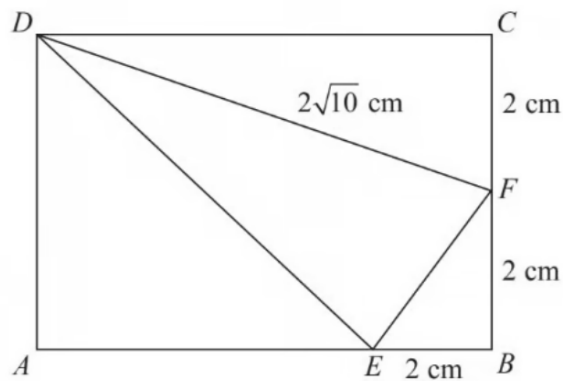


Diagram **NOT**
accurately drawn

Show that the area of triangle DEF is 8 cm^2

You must show all your working.

(4 marks)

Q38**PRACTICE**

Practice · Very Hard

CH 4 · TOPIC 10

5 marks

- 1 The diagram represents a metal frame.

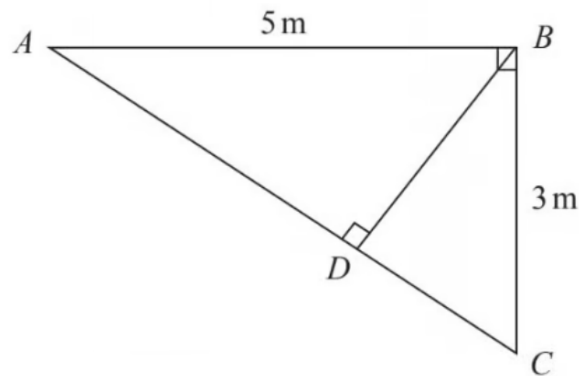


Diagram **NOT**
accurately drawn

The frame is made from four metal bars, AB , AC , BC and BD .

Angle $ABC = \text{angle } ADB = 90^\circ$

$AB = 5 \text{ m}$

$BC = 3 \text{ m}$

Work out the total length of the four metal bars of the frame.

Give your answer correct to 3 significant figures.

(5 marks)

Q39**PRACTICE**

Practice • Very Hard

CH 4 • TOPIC 10 5 marks

10 (a) Without using a calculator, show that $\sqrt{20} = 2\sqrt{5}$.

(2 marks)

(b) The point A is shown on the unit grid below.

The point B is $2\sqrt{5}$ units from A and lies on the intersection of two grid lines.

Mark **one** possible position for B.



(3 marks)

PAST PAPERS

Q40**PAST PAPER**

Edexcel 4MA1 · January 2020 · Paper 1HR · Q15

CH 4 · TOPIC 10 **4 marks**

15 The diagram shows two right-angled triangles, DEF and EFG .

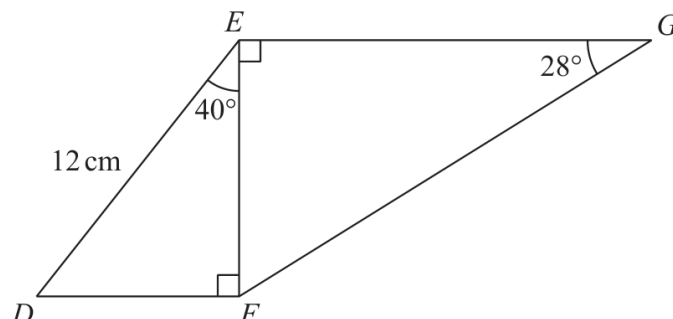


Diagram **NOT**
accurately drawn

Work out the length of EG .

Give your answer correct to 3 significant figures.

..... cm

(Total for Question 15 is 4 marks)

Q41**PAST PAPER**

Edexcel 4MA1 · June 2022 · Paper 2HR · Q10

CH 4 · TOPIC 10 **5 marks**

10 The diagram shows triangle ABP inside the regular hexagon $ABCDEF$

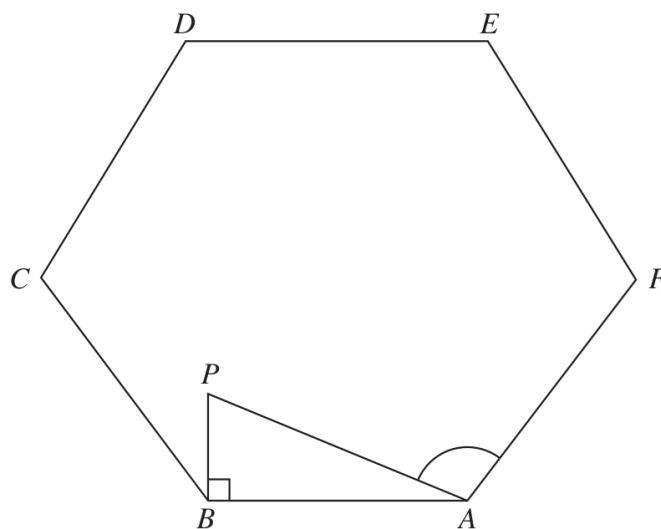


Diagram **NOT**
accurately drawn

$$AB = 5 \text{ cm}$$

$$BP = 2 \text{ cm}$$

$$\text{Angle } ABP = 90^\circ$$

Work out the size of angle PAF

Give your answer correct to 3 significant figures.

○

(Total for Question 10 is 5 marks)

Q42**PAST PAPER**

Edexcel 4MA1 · June 2023 · Paper 2H · Q12

CH 4 · TOPIC 10 **4 marks**

12 The diagram shows right-angled triangle ABD

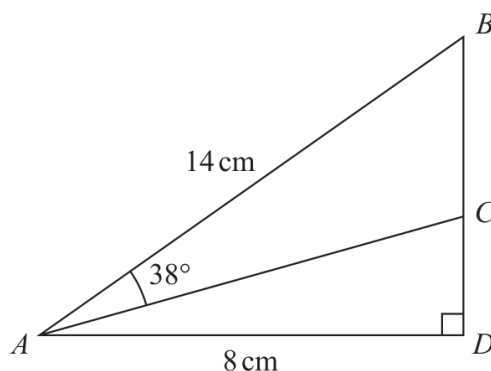


Diagram **NOT**
accurately drawn

$$AB = 14 \text{ cm} \qquad AD = 8 \text{ cm}$$

C is the point on BD such that angle $BAC = 38^\circ$

Work out the length of CD

Give your answer correct to 3 significant figures.

..... cm

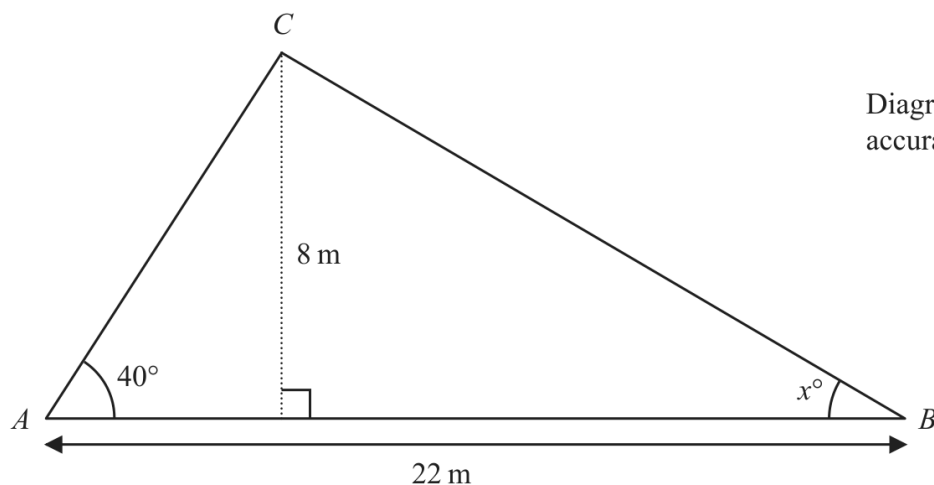
(Total for Question 12 is 4 marks)

Q43**PAST PAPER**

Edexcel 4MA1 • November 2023 • Paper 1H • Q10

CH 4 • TOPIC 10**5 marks**

10 Here is triangle ABC



Work out the value of x
Give your answer correct to one decimal place.
Show your working clearly.

 $x = \dots\dots\dots$

(Total for Question 10 is 5 marks)

Q44**PAST PAPER**

Edexcel 4MA1 • November 2023 • Paper 2H • Q2

CH 4 • TOPIC 10 4 marks

- 2 The diagram shows triangle ABC and triangle ECD

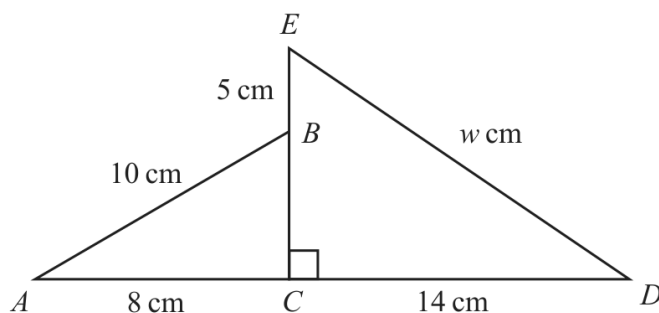


Diagram **NOT**
accurately drawn

ACD and EBC are straight lines.

$$AB = 10 \text{ cm} \quad AC = 8 \text{ cm} \quad EB = 5 \text{ cm} \quad CD = 14 \text{ cm} \quad ED = w \text{ cm}$$

Work out the value of w

Give your answer correct to one decimal place.

$$w = \dots\dots\dots$$

(Total for Question 2 is 4 marks)

Q45**PAST PAPER**

Edexcel 4MA1 • November 2024 • Paper 1H • Q12

CH 4 • TOPIC 10 4 marks

- 12 ABC is a right-angled triangle.
 D is a point on BC

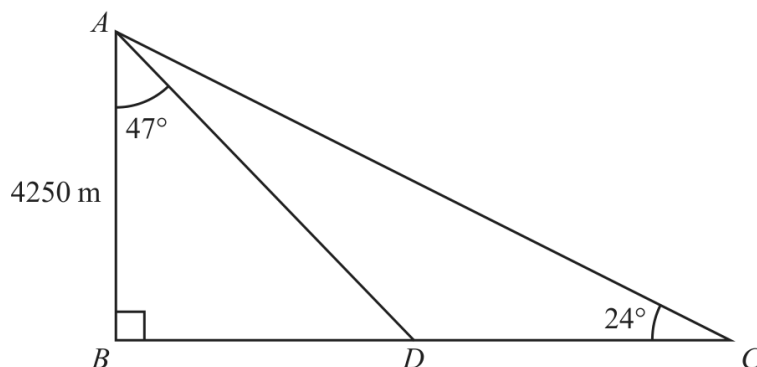


Diagram **NOT**
accurately drawn

$$AB = 4250 \text{ m} \quad \text{angle } BAD = 47^\circ \quad \text{angle } BCA = 24^\circ$$

Work out the length of DC

Give your answer correct to the nearest integer.

..... m

(Total for Question 12 is 4 marks)

Q46**PAST PAPER**

Edexcel 4MA1 • June 2025 • Paper 2HR • Q11

CH 4 • TOPIC 10 5 marks

11 The diagram shows two triangles, ADE and CDB

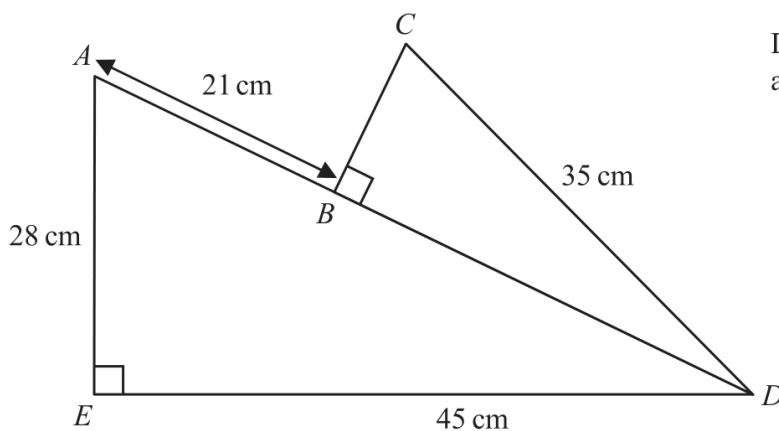


Diagram **NOT**
accurately drawn

ABD is a straight line.

$AE = 28 \text{ cm}$ $ED = 45 \text{ cm}$ $AB = 21 \text{ cm}$ $CD = 35 \text{ cm}$

angle $AED = \text{angle } CBD = 90^\circ$

Work out the area of triangle CDB

Give your answer correct to 3 significant figures.

..... cm^2

(Total for Question 11 is 5 marks)

Q47**PAST PAPER**

Edexcel 4MA1 · June 2025 · Paper 2H · Q10

CH 4 · TOPIC 10 5 marks

10 ABD and ABC are right-angled triangles.

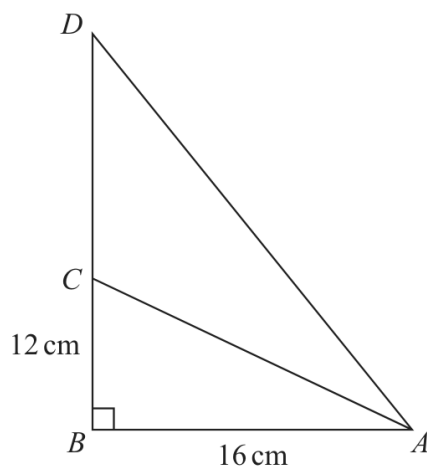


Diagram **NOT**
accurately drawn

$$AB = 16 \text{ cm} \quad BC = 12 \text{ cm} \quad AD = 1.5 \times AC$$

Find the length of CD

Give your answer correct to 3 significant figures.

..... cm

(Total for Question 10 is 5 marks)

Q48**PAST PAPER**

Edexcel 4MA1 • November 2025 • Paper 1H • Q11

CH 4 • TOPIC 10 4 marks

11 ABC is a right-angled triangle.

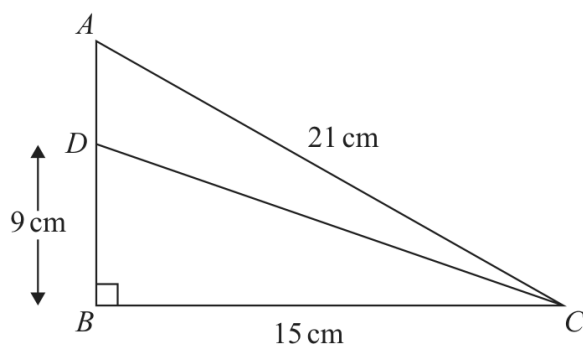


Diagram **NOT**
accurately drawn

$AC = 21 \text{ cm}$ $BC = 15 \text{ cm}$ angle $ABC = 90^\circ$

The point D lies on AB such that $DB = 9 \text{ cm}$

Work out the size of angle ACD

Give your answer correct to one decimal place.

o

(Total for Question 11 is 4 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|----------------------|---|
| 01 | Pythagoras | Two sides, no angle; add for the hypotenuse, subtract for a shorter side. |
| 02 | SOH CAH TOA | Label from the angle; inverse to find an angle. |
| 03 | Elevation | Draw the horizontal; elevation = depression. |
| 04 | Two triangles | Shared side first; no early rounding. |

BOTTOM LINE Right angle? → two sides: Pythagoras • side and angle: SOH CAH TOA.

Answers

Right-Angled Triangles: Pythagoras and Trigonometry • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • PYTHAGORAS

Q1	Dr Eslam • Hypotenuse	$\sqrt{49 + 576} = 25 \text{ cm}$
Q2	Dr Eslam • Shorter side	$\sqrt{169 - 25} = 12 \text{ cm}$
Q3	Practice • Easy	3.52 m
Q4	Practice • Medium	slant $\sqrt{149} = 12.2$: 51.2 m > 50: No
Q5	Practice • Medium	$MN = 10.3$: 32.4 cm
Q6	Practice • Very Hard	diagonal = diameter: $x = \sqrt{\frac{98}{\pi}} = 5.59$
Q7	Edexcel 4MA1 • January 2020 • Paper 2HR • Q4	10.2 cm
Q8	Edexcel 4MA1 • November 2021 • Paper 1H • Q2	$h = 42.5$
Q9	Edexcel 4MA1 • January 2022 • Paper 1HR • Q3	legs 12.8 and 9.6: $16 \times 3 + 3.2 + 9.6 \times 2 = 70.4$ cm
Q10	Edexcel 4MA1 • May 2023 • Paper 1H • Q5	$AB = 9$: 33.6 cm
Q11	Edexcel 4MA1 • May 2024 • Paper 1H • Q11	$MC = 6.71 \rightarrow 7 \text{ m}$; $12 + 9 + 9 + 7 = 37 \text{ m}$: 795.50 euros
Q12	Edexcel 4MA1 • November 2025 • Paper 2H • Q11	$AC = 16, BC = 34$: 80 cm

IDEA 02 • SOH CAH TOA

Q13	Dr Eslam • Find a side	$10 \sin 35^\circ = 5.74 \text{ cm}$
Q14	Dr Eslam • Find an angle	$\tan^{-1} \frac{5}{12} = 22.6^\circ$
Q15	Practice • Easy	$\cos^{-1} \frac{6.4}{9.6} = 48.2^\circ$
Q16	Practice • Easy	$16 \sin 38^\circ = 9.85 \text{ cm}$
Q17	Practice • Medium	$2 \times 8 \sin 70^\circ = 15.0 \text{ cm}$
Q18	Practice • Hard	$7 \times 5 \sin 40^\circ = 22.5 \text{ cm}^2$
Q19	Edexcel 4MA1 • January 2020 • Paper 1H • Q9	53.5 cm
Q20	Edexcel 4MA1 • January 2021 • Paper 1H • Q10	$AB = 17.65, AD = 7.46: 50.2 \text{ cm}$
Q21	Edexcel 4MA1 • May 2021 • Paper 2H • Q7	$x = 9.7$
Q22	Edexcel 4MA1 • January 2022 • Paper 1H • Q3	7.61 cm
Q23	Edexcel 4MA1 • June 2022 • Paper 2H • Q6	$x = 7.1$
Q24	Edexcel 4MA1 • June 2024 • Paper 2H • Q5	$x = 4.4$
Q25	Edexcel 4MA1 • November 2024 • Paper 2H • Q3	$x = 6.3$
Q26	Edexcel 4MA1 • June 2025 • Paper 1HR • Q9	14.6 cm

IDEA 03 • ELEVATION, DEPRESSION AND REAL LIFE

- Q27** Dr Eslam • Elevation $40 \tan 32^\circ = 25.0 \text{ m}$
- Q28** Practice • Medium $\cos y = \frac{2.25}{6}$: $y = 68.0^\circ < 75^\circ$: not safe
- Q29** Practice • Medium (a) 7.5 km (b) $180 + 36.9$: 217°
- Q30** Practice • Hard $CD = 2.55$: rise 0.85 m: $BD = 0.664 \text{ m}$
- Q31** Practice • Very Hard (a) $h = 2 + 15 - 15 \cos x$ (b) 24.5 m (c) $P(180, 32)$
- Q32** Edexcel 4MA1 • January 2021 • Paper 2HR • Q13 195.4 m
- Q33** Edexcel 4MA1 • November 2021 • Paper 1H • Q12 (a) $d = 275$ (b) 28.1 m
- Q34** Edexcel 4MA1 • January 2022 • Paper 2HR • Q12 5.7°
- Q35** Edexcel 4MA1 • January 2022 • Paper 2H • Q14 3.65 m

IDEA 04 • TWO TRIANGLES TOGETHER

- Q36** Practice • Hard $BD = \sqrt{75}$: $x = \sqrt{91} = 9.54$
- Q37** Practice • Hard $DC = 6$: $24 - 8 - 6 - 2 = 8 \text{ cm}^2$
- Q38** Practice • Very Hard $AC = \sqrt{34}$, $BD = \frac{15}{\sqrt{34}}$: 16.4 m
- Q39** Practice • Very Hard (a) $\sqrt{4 \times 5}$ (b) 4 right and 2 up (or 2 and 4)
- Q40** Edexcel 4MA1 • January 2020 • Paper 1HR • Q15 $EF = 9.19$: $EG = 19.6 \text{ cm}$
- Q41** Edexcel 4MA1 • June 2022 • Paper 2HR • Q10 $120 - 21.8 = 98.2^\circ$
- Q42** Edexcel 4MA1 • June 2023 • Paper 2H • Q12 $CD = 2.47 \text{ cm}$
- Q43** Edexcel 4MA1 • November 2023 • Paper 1H • Q10 $AD = 9.53$: $x = 32.7$
- Q44** Edexcel 4MA1 • November 2023 • Paper 2H • Q2 $BC = 6$, $EC = 11$: $w = 17.8$
- Q45** Edexcel 4MA1 • November 2024 • Paper 1H • Q12 4988 m
- Q46** Edexcel 4MA1 • June 2025 • Paper 2HR • Q11 $AD = 53$, $BD = 32$, $CB = 14.2$: 227 cm^2
- Q47** Edexcel 4MA1 • June 2025 • Paper 2H • Q10 $AC = 20$, $AD = 30$, $BD = 25.4$: $CD = 13.4 \text{ cm}$
- Q48** Edexcel 4MA1 • November 2025 • Paper 1H • Q11 $AB = \sqrt{216} = 14.7$: $44.4 - 31.0 = 13.5^\circ$